

BUILDING A WINNER FROM SCRATCH

A Comprehensive NBA Expansion Draft Strategy

CMU SPORTS ANALYTICS CLUB × SIS COMPETITION | APRIL 2026

Our Journey

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The Problem & Data

190 available players, 4 data sources, 13 play types

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Modeling Journey

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The EAP Metric

Our novel metric: balancing theoretical, tangible, team impacts

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Team A: First pick in expansion draft and 7th overall pick

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5 high EAP with low usage

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30-player board, 15-player simulated draft

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Payroll, cap space, multi-year trajectory

What We Were Given

190

Available Players

13

Offensive Play Types

13

Defensive Play Types

52

Total Data Files

Offense Files

13 play types · PPC, Exp PPC, EPI, PPC+

Defense Files

13 play types · on-ball defensive metrics

Advantages Created

Adv Crt, Adv Crt+ · offensive creation rate

Adv Prevented

Adv Prev, Adv Prev+ · defensive disruption rate

5-Man Lineups

17K+ combos · EPI, PPC+ offense & defense

Available Players

Salary data through 2029-30, 30 teams

Approach 1: XGBoost on Lineup Data

The Attempt

Input: 17,588 five-man lineups with PPC, Exp PPC, EPI, PPC+

Target: Predict lineup metrics from individual player metrics

Result: $R^2 = 0.48$ (train) / 0.05 (test) — massive overfit

WHY IT FAILED

Extreme feature sparsity: 17K+ lineup combinations $\times$ 558 players = model memorized noise. Some lineups appeared fewer than 5 times — there is simply not enough signal for generalization.

Lesson Learned

■ Lineup-level prediction needs large, dense data

■ Sparse matrices amplify noise, not signal

■ Shift focus to player-level evaluation

■ Individual metrics are more reliable predictors

Approach 2: Linear Regression on Advantage Metrics

The Attempt

Input: Adv Crt, Adv Prev, Adv Allow, Adv Rdce across all 13 play types

Target: Linear regression predicting player contribution score

Finding: 8 of 12 advantage metrics heavily right-skewed
Power transforms applied — distributions still non-normal

WHY IT FAILED

Linear regression assumes normally distributed residuals. Even after Box-Cox / Yeo-Johnson power transformations, skewness persisted. Forcing non-normal data through OLS produces biased, unreliable estimates.

Lesson Learned

■ High-usage players dominate raw advantage counts

■ Volume bias can't be fixed by transformation alone

■ Need a ratio metric, not a regression model

■ Build a custom framework directly from the data

Our Metric **EAP** Value = EPI + Advantage Ratio + PPC Ratio

Combine Theoretical, Tangible, and Team Impact Metrics

$$\begin{array}{ccc} \text{Min-Max} & \text{Min-Max} & \text{Min-Max} \\ \text{Normalization} & \text{Normalization} & \text{Normalization} \\ \left(\text{Net EPI} \right) & + & \left(\frac{\text{Offensive PPC}}{\text{Defensive PPC}} \right) & + & \left(\frac{\text{Advantages Created} + \text{Advantages Prevented}}{\text{Advantages Lost} + \text{Advantages Allowed}} \right) \\ .5 & & .25 & & .25 \end{array}$$

EAP: Top Overall Players

Player	Metric Scaled
Shai Gilgeous-Alexander	0.725647
Donovan Mitchell	0.499803
Kawhi Leonard	0.493801
Tyrese Maxey	0.484059
Jalen Brunson	0.482912
Kevin Durant	0.453721
James Harden	0.427209
Devin Booker	0.388564
Jamal Murray	0.385325
Luka Doncic	0.38328
Derrick White	0.381441
Jaylen Brown	0.378566
Anthony Edwards	0.376117
Stephen Curry	0.374543
Brandon Ingram	0.364301

- Validation by basketball intuition
- Favors Guards
- Slightly favors offensive metrics (but so does the NBA)



5 Hidden Gems

Process

Find Players with High EAP Scores, Low Opportunities (with a minimum sample threshold)



Results

Player	Percent difference in EAP percentile and usage percentile
Isaiah Stevens SAC	77%
Cameron Payne PHI	59%
Pat Connaughton CHA	48%
D'Angelo Russell DAL	43%
Cormac Ryan MIL	41%

We Choose: Team A



6th Overall Pick:
Top 0.459 Percentile

7th Overall Pick:
Top 0.542 Percentile

1. Expansion Draft Value

We estimate that Team A's starting five and top nine players metrics are around 4% more than team B's.

2. Rookie Picks are Around Equal

Since the bubble, according to our metrics, the sixth pick marginally underperforms the seventh pick consistently. This year seems to hold that same trend with Keaton Wagler and Darius Acuff Jr.

3. Max Metric Correlation

Analysis reveals that the best player on a team significantly contributes to overall team performance.

30-Player Big Board: Draft Philosophy

Top 2 Players/Team

Uses EAP metrics, filters low samples.

Opportunities Factor

If two players have within 10% EAP value and larger than 10% opportunities difference, take player with more experience

Player Salary Cap

For low-level roster players, do not sign players with more than 10 million dollar contracts

Minimize Cap Hit

We want to target free agents and compete in the years to come, not necessarily the first year. We aim for value contracts, especially as a franchise with many more years to grow.

RFA STRATEGY:

62 of 190 available players are Restricted Free Agents. We keep this in mind in the financial analysis section.

Our Draft Picks - Shiny App

Pick	Player	Team	2026–27	2027–28	2028–29	2029–30	Status
1	Cameron Payne	PHI	RFA	—	—	—	RFA
3	Jerami Grant	POR	\$34,206,898	\$36,413,790	—	—	Roster
5	D'Angelo Russell	WAS	\$5,969,250	—	—	—	Roster
7	Al Horford	GSW	\$5,969,250	—	—	—	Roster
9	Dru Smith	MIA	\$2,584,539	\$2,934,742	—	—	Roster
11	Brook Lopez	LAC	\$9,187,500	—	—	—	Roster
13	Luke Kornet	SAS	\$10,450,000	\$9,900,000	\$9,350,000	—	Roster
15	Corey Kispert	ATL	\$13,975,000	\$13,025,000	—	—	Roster
17	Pat Connaughton	CHA	\$3,815,861	—	—	—	Roster
19	Goga Bitadze	ORL	\$7,608,696	—	—	—	Roster
21	Killian Hayes	SAC	\$3,018,158	—	—	—	Roster
23	Jaylen Clark	MIN	RFA	—	—	—	RFA
25	Pacome Dadiet	NYK	\$2,983,680	\$5,373,608	—	—	Roster
27	Elijah Harkless	UTA	RFA	—	—	—	RFA
29	Chaney Johnson	BKN	RFA	—	—	—	RFA

NBA Draft 7th Overall Pick - **Darius Acuff Jr.** *ARK*

Via The Athletic

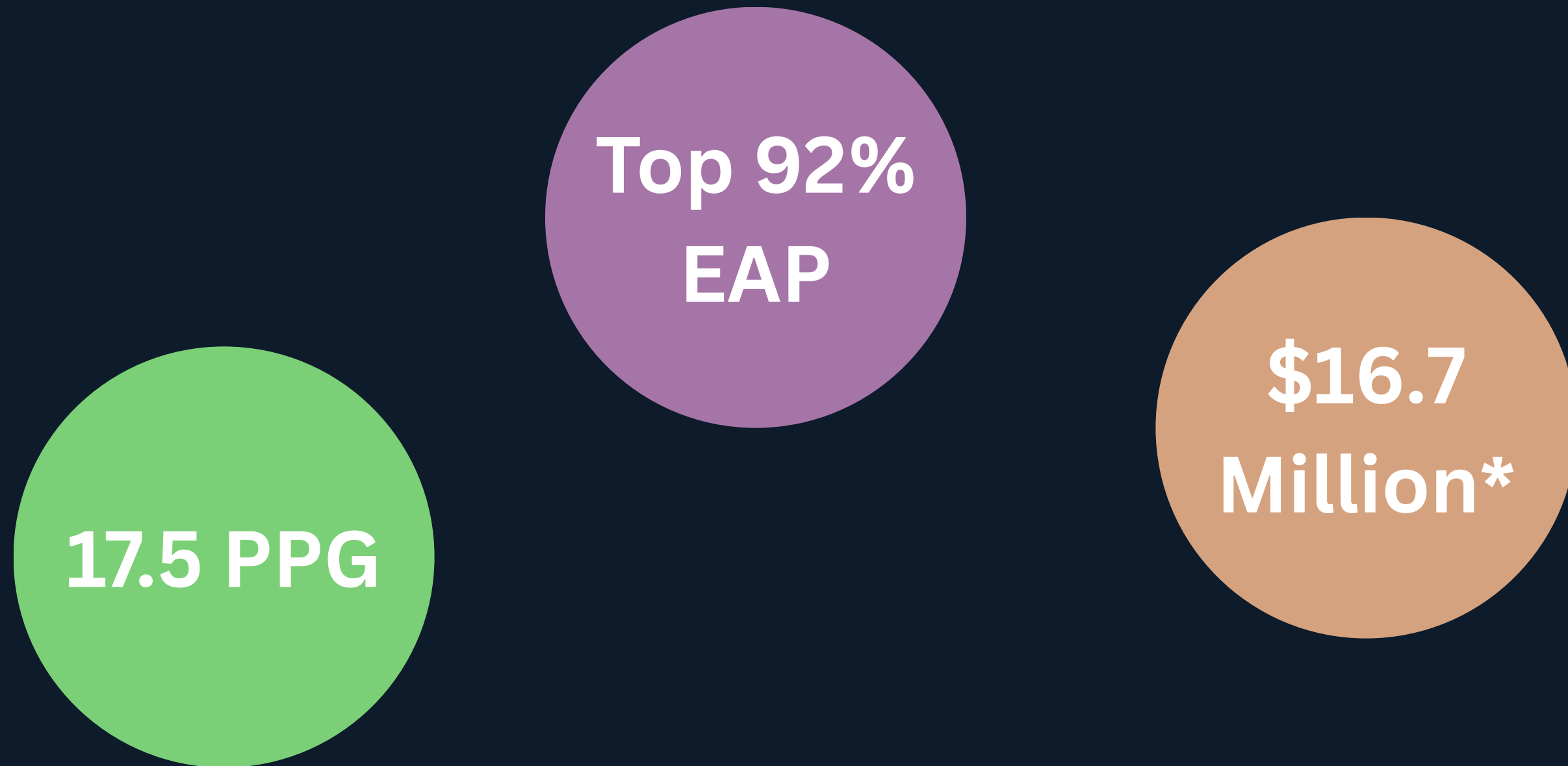
22 PPG

6.2 APG

42.9%
3FG

49.1%
FG

Free Agency Pickup: **Coby White** CHA



*20% more base than his last contract, 5% raises throughout

Year 1 Payroll Summary (2026-2027)

Player	2026-27 Salary
Jerami Grant	\$34,206,898
Coby White	\$15,466,667
Corey Kispert	\$13,975,000
Luke Kornet	\$10,450,000
Brook Lopez	\$9,187,500
Goga Bitadze	\$7,608,696
#7 overall pick 2026: Projected Darius Acuff	\$6,705,400
D'Angelo Russell	\$5,969,250
Al Horford	\$5,969,250
Pat Connaughton	\$3,815,861
Cameron Payne	\$3,750,000
Killian Hayes	\$3,018,158
Pacome Dadiet	\$2,983,680
Dru Smith	\$2,584,539
#37 pick 2026	\$1,400,327
TOTAL PAYROLL	\$127,091,226
SALARY CAP Y1	\$109,263,850

Starting Five - Year 1



Rotation

- D'Angelo Russell** (G)
- Brook Lopez** (C)
- Jerami Grant** (F)
- Goga Bitadze** (PF/C)

Team Strengths & Playing Identity

OFFENSIVE IDENTITY

Smaller Guards Play off of Tall F/C

Both starting our starting center and power forward are above even foot tall.

White and Payne Are Initiators

3.48 and 12.00 Adv Crt, respectively, for initiation with greater than 50 sample size.

Horford, Kornet P&R

Pair up our guards with the bigs, look to lob toward basket.

Acuff Volume 3s

42.9% from three point range (will likely drop at least a little from college to NBA).

DEFENSIVE IDENTITY

Man Defense, Minimizing Switching

Our lineup has an array of sizes, so we want to minimize height mismatches.

Help Defense

Russell and White are rather weak in isolation, so we need to provide support.

Force Half Court Offense

Russell, Horford, and Kornet are high-level at transition defense, according to PPC.

Deny Rim Access

Payne, Horford, and Kornet excel at gather defense.

Starting Five Predicted Performance*

Top 82%

based on EAP

Top 85%

based on net EPI

Top 52%

based on PPC Ratio

Lineup already has just under the average number of opportunities, so the lineup should scale nicely.

*Relative to other high-volume lineups

2026-27 Season Projection

BEST CASE

44-50 Wins

- Coby White scales to a high-level point guard
- Darius Acuff Jr. continues to shoot the three well at the NBA line
- The small guards and tall centers/forwards mesh extremely well, creating more spacing advantages

BASE CASE

38-44 Wins

- Chemistry is moderate, players adjust to new team well
- Team doesn't have a consistent 25-26 points per game player
- Rookie draft pick does well in his role

WORST CASE

32-38 Wins

- Players have a hard time adjusting to more volume
- Team fails to generate homecourt advantage
- Small guard size could perform poorly against larger lineups
- We fail to attract a good head coach

PROJECTED WIN RANGE

37 – 45

WINS

Play-In → 6 Seed

*Based on rank of EAP, EPI, PPC Metrics from previous slide

Year 2 Payroll Summary (2027-2028)

Player	2027-28 Salary
Jerami Grant	\$36,413,790
Coby White	\$16,240,000
Corey Kispert	\$13,025,000
Luke Kornet	\$9,900,000
Brook Lopez	\$9,646,875
Goga Bitadze	\$7,989,131
#7 overall pick 2026: Projected Darius Acuff	\$7,040,800
Marcus Smart	\$6,468,840
Pacome Dadiet	\$5,373,608
#12 overall pick 2027	\$5,325,500
Cameron Payne	\$3,937,500
Pat Connaughton	\$3,500,000
Dru Smith	\$2,934,742
#37 pick 2026	\$2,193,935
#42 overall pick 2027	\$1,330,311
TOTAL PAYROLL	\$131,320,032
SALARY CAP Y2	\$138,958,143



Starting Five - Year 2



Rotation	
G	Marcus Smart
C	Brook Lopez
F	Jerami Grant
G	Dru Smith
G/F/C	12th Pick 2027

Year 3 Payroll Summary (2028-2029)

Player	2028-29 Salary
Deni Advija	\$42,000,000
Jerami Grant	\$20,000,000
Duncan Robinson	\$18,181,466
Coby White	\$17,052,000
Luke Kornet	\$9,350,000
Goga Bitadze	\$8,388,587
#7 overall pick 2026: Projected Darius Acuff	\$7,375,800
Marcus Smart	\$6,792,282
#12 overall pick 2027	\$5,592,000
#18 overall pick 2028	\$4,306,300
Cameron Payne	\$4,134,375
Pat Connaughton	\$3,675,000
Dru Smith	\$3,284,945
#42 overall pick 2027	\$1,356,917
#48 overall pick 2028	\$1,303,705
TOTAL PAYROLL	\$152,793,378
SALARY CAP Y3	\$184,085,794

Top 96%
EAP

~25 PPG

Starting Five - Year 3



Rotation	
G	Cameron Payne
C	Duncan Robinson
F	Jerami Grant
G	Marcus Smart
G/F/C	Dru Smith/ 12th Pick 2027

Year 4 Payroll Summary (2029-2030)

Player	2029-30 Salary
Jalen Brunson/Derrick White/Tyrese Maxey	\$60,000,000
Deni Advija	\$45,360,000
Duncan Robinson	\$19,090,540
Coby White	\$17,904,600
Luke Kornet	\$11,220,000
#7 overall pick 2026: Projected Darius Acuff	\$9,367,266
Goga Bitadze	\$8,808,017
Marcus Smart	\$7,131,896
#12 overall pick 2027	\$5,858,300
#18 overall pick 2028	\$4,521,200
Cameron Payne	\$4,465,125
Pat Connaughton	\$3,858,750
#23 overall pick 2029	\$3,842,200
Dru Smith	\$3,449,192
#48 overall pick 2028	\$1,329,779
TOTAL PAYROLL	\$207,406,273
SALARY CAP Y3	\$195,095,177

Maxey:
5th in
EAP

Brunson:
6th in
EAP

Starting Five - Year 4



Rotation

- Duncan Robinson** (SF)
- Goga Bitazde** (C)
- Cameron Payne** (G)
- Marcus Smart** (G)
- Dru Smith/12th Pick 2027** (G/F/C)

Four-Year Competitive Trajectory

Year 2
2027-28

Aspiring Play-in Team

34-48 Wins

Focus on continuing to build around Acuff and Coby White by adding defensive depth through Marcus Smart and the draft. Expect slight stagnation due to Horford retirement, will a target a high-EAP free-agent (Deni Advija)

Cap: \$138M committed, \$7M over Y2 cap

Moves Made: Brought up Marcus Smart

 6 Seed — Play In

Year 3
2028-29

Competitive, but still missing a piece

44-54 Wins

Advija leads the team to playoffs, while there is a still a missing piece. Coby White showcases playmaking ability with big men and shooting (Duncan Robinson)

Cap: \$152.7M Committed, 32M in outright cap space

Moves Made: Brought in Deni Advija, Duncan Robinson

 4-6 Seed — Playoff Run

Year 4
2029-30

Championship Window

54-62 Wins

Starting 5 is projected to have the best net EPI in the league with any of Brunson, White, or Maxey. They'd be also be the best in terms of EAP with a strong bench of Duncan Robinson, Cam Payne, and Dru Smith

Cap: \$207M committed

Moves Made: Big splash for a star (top 10 in EAP)

 Contender — Top 4 Seed



BUILDING A WINNER FROM SCRATCH

Carnegie Mellon University

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